

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)			Diagram should show: Polarisation of N—H or O—H bonds in 2-aminoethanol (1) Lone pairs of electrons used in hydrogen bonding to nitrogen or oxygen atoms (1) Polarisation of water molecules (1) Hydrogen bonds indicated between 2-aminoethanol and water molecules using relevant nitrogen / oxygen and hydrogen atoms (1) If no water then 3 marks maximum	2	2		4		
	(b)			The forces of attraction between molecules of 2-aminoethanol are stronger than the attractive forces between molecules of 1,2-diaminoethane (as the former has a higher boiling temperature) (1) This suggests that intermolecular hydrogen bonding between / involving O and H is stronger than the hydrogen bonding between N and H (1) This is (probably) due to a greater electronegativity difference between O and H than between N and H / O more electronegative than N / size considerations (1)		1	2	3		